01-11-2026

Agenda:

1) → How python executes work
2) → Threading
3) → Multiprocess
4) → pool executor
5) → asyncio
6) → linux env installation (optional)
7) → FDE concept (AI RMF) (compact)

Announcement:

7th & 8th November

Holiday IST

next weekend

level: Hard

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core → 12

logical core → 24 (hardware)

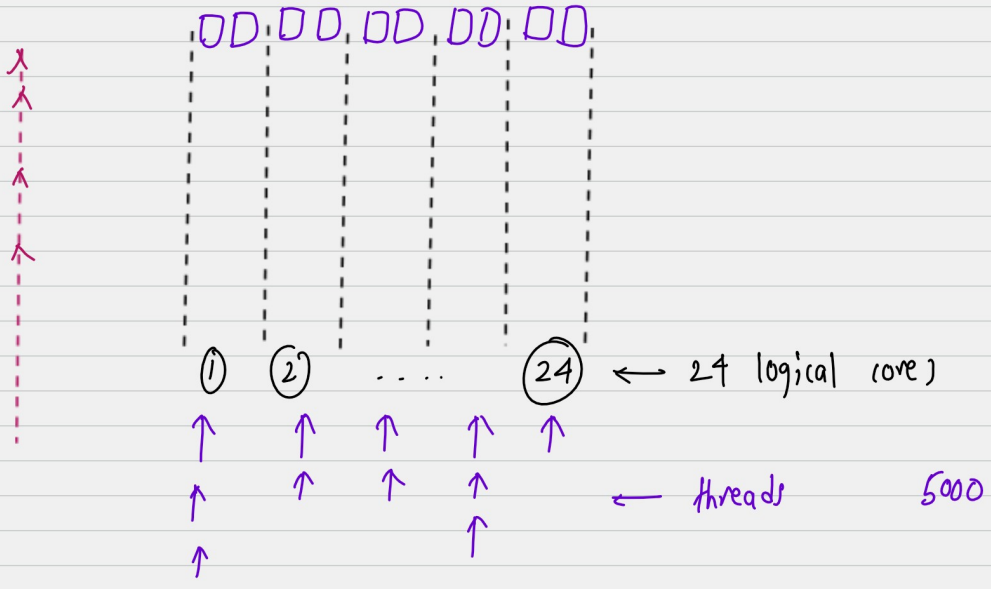
1 core → 1 to 2 threads

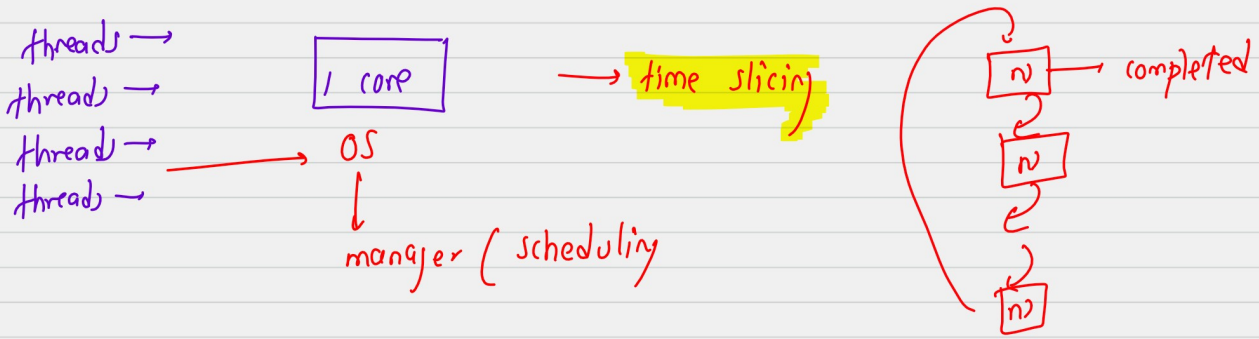
Airport

process → (chrome, Discord, whatsapp)

small task

airline (process)





python → GIL → 1 thread (task)

OS, time slicing

↳ Threading → multiple threads → T1 → Task, T2 → Task, T3 → Task

↳ multiprocess → task → complex → prime number (1, 100,000) → CPU intensive → 2 sec

P1 → P_n (1, 10,000)
P2 → P_n (10,000, 50,000)
P3 → P_n (50,000, 100,000)

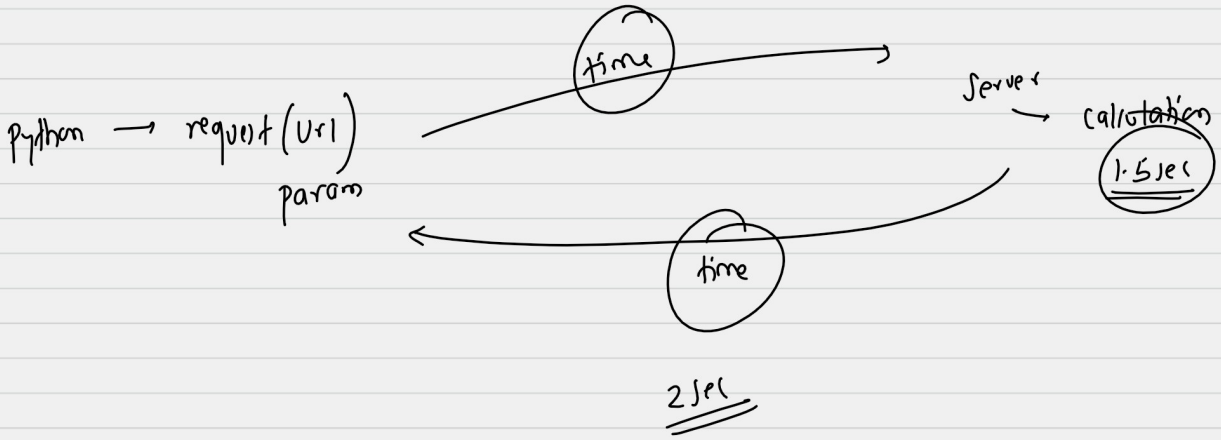
skip → GIL

P1 P2 P3 → (core)

↳ Asyncio (library)

↳ different scheduling mechanism (not OS time slicing)

↳ one thread → task, task ← manage (event loop)



I/O Bound work
-> calling an api/url
-> Reading from database
-> Downloading
-> Uploading

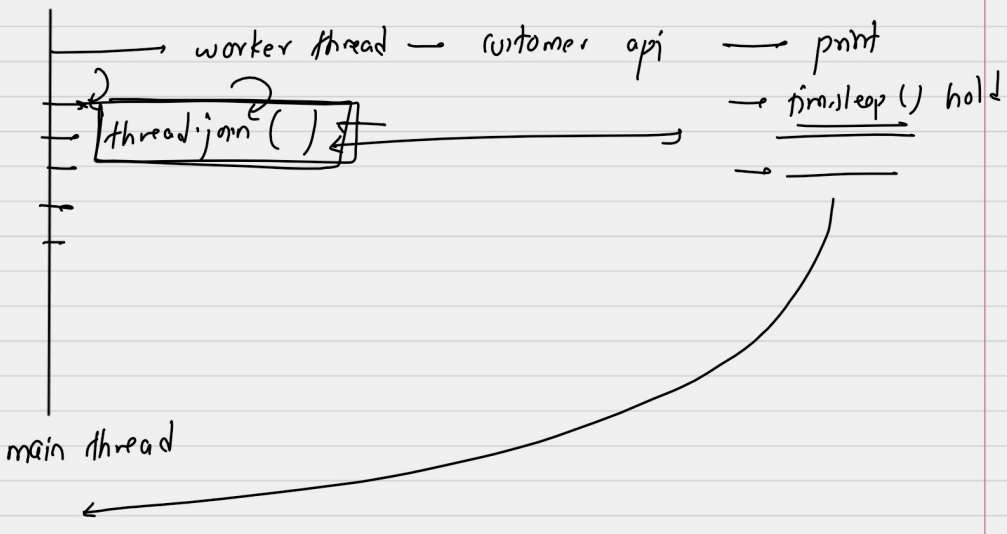
CPU Bound work
-> Process large arrays in python
-> Mathematical result
-> Video / image based operation

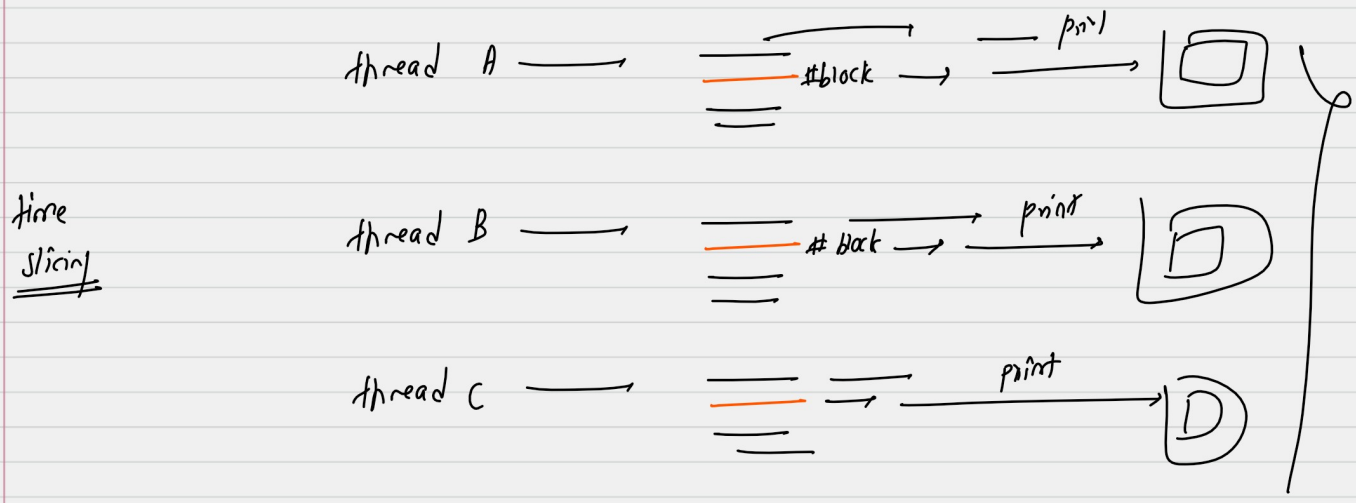
kitchen

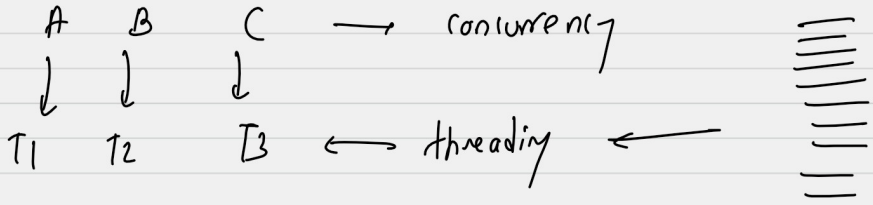
- prepare cuistor order (A)
- prepare payment receipt (B)
- check inventor (C)

sharing → resource of rest

running a restaurant → process → task, task, task

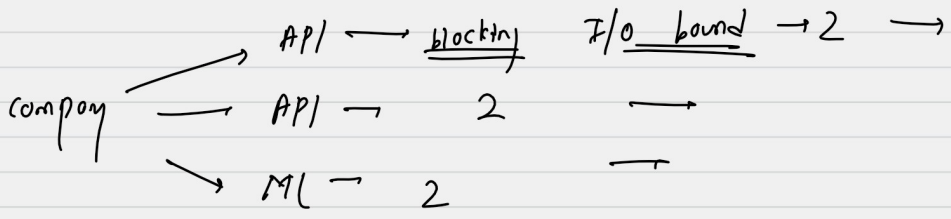






T1 manager (event loop) → Asyncio

Share memory



Thread	counter	current	waiting
A	0	0	start B
B	0	0	start A
A	1	0	time slicing
B	1	0	

Monitoring company

customer API (2) → T1

payment API (2) → T2

Inventory API (2) → T3

6 — 2 + overhead

company → uploads → 500 images from its cameras → AI (0.2s/image) → detect helmet, PPE kit

5 sec → 300 × 0.2 → 100 seconds

12:00:00 → 500 → 100 seconds

+5sec

12:00:05 → 500 → 95 + 100

+5

12:00:10 → 500 → 90 + 95 + 100

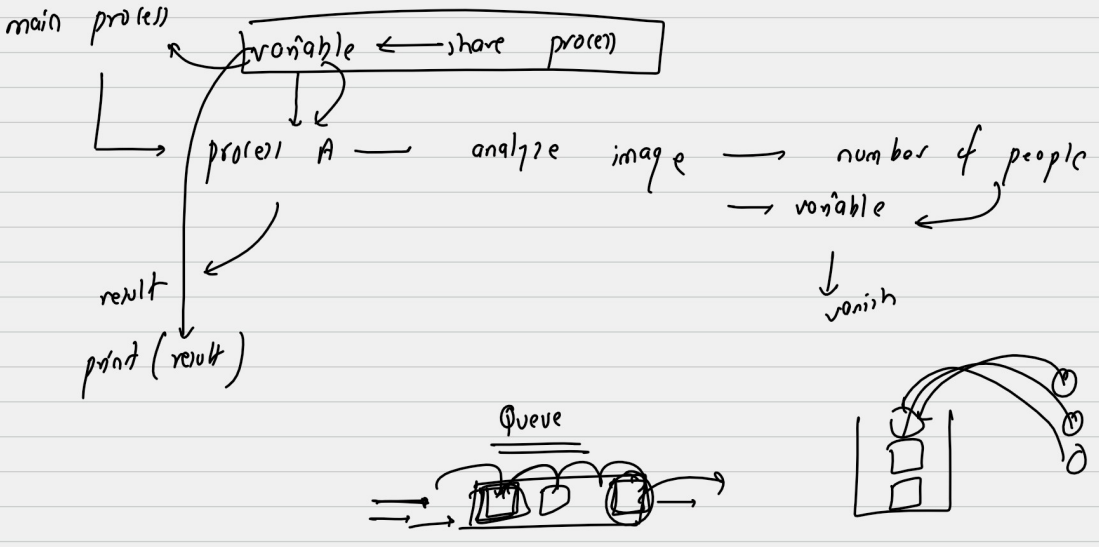
Threads? → NO
→ GIL
→ threads → I/O bound
→ → cpu bound → wait → complex

→ execution model → multiprocessing

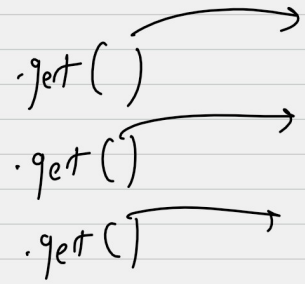
process → running instance of a program
→ multiple process do not share executing environment
→ Different memory space

GIL → process

process → core



First in - First out (FIFO)



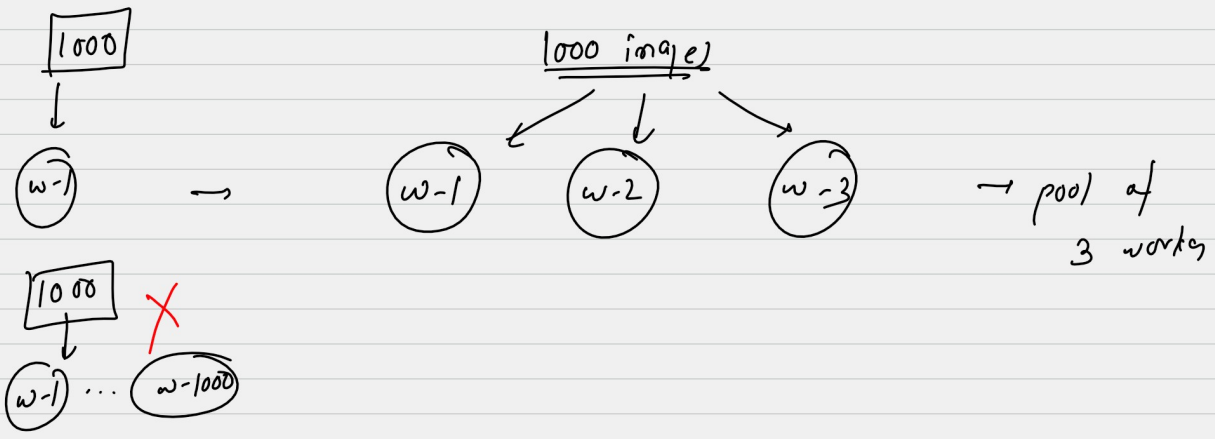
reciving → 1000 of image

→ manuall, proces → assign work

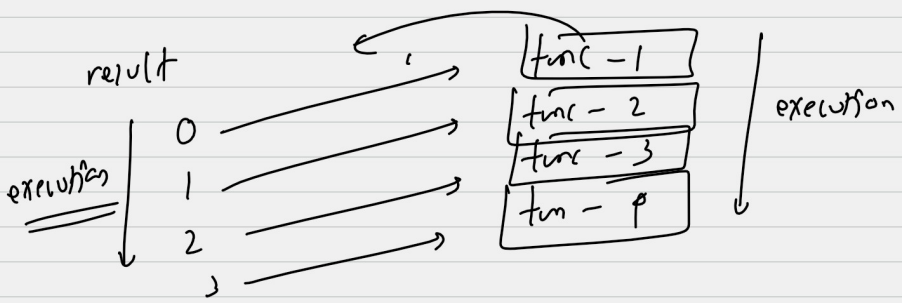
small task

- start
- wait
- collect

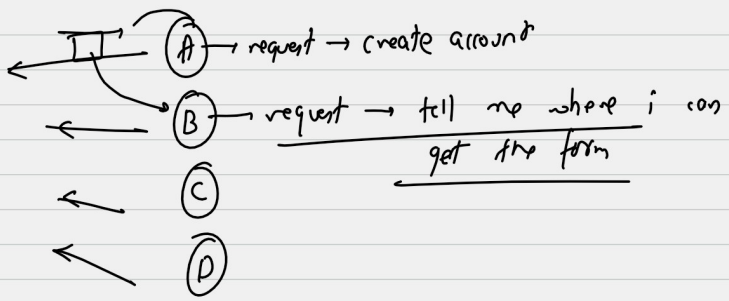
executor & proces pool



result = map(func, [1,2,3,4])



problem → Bank manager



result
↓
B

order → status → added to cart
→ payment done
→ order placed
→ dispatched
→ delivered

future → eventual outcome of submitted job
submit ()
→ waiting to start
→ running
→ finished
→ finished with exception
→ cancelled before it started.

FDE

AI-RMF

NIST AI Risk Management Framework

NIST : National Institute of Standards and Technology
RMF : Risk Management Framework

ISO -> standards (certified)
NIST RMF -> Guidance (January 26, 2023) -> latest update > (April 7, 2026)

NIST AI RMF is a voluntary framework that helps organizations incorporate trustworthiness and risk management into the design, development, deployment, use, and evaluation of AI systems.

use :— computer vision solution

Business (Developing) (proving the service) ← Client

industrial safety
↳ helmet
↳ PPE kit
↳ entering in the restricted zone



↳ analytics (video)
↓
monitoring platform

NIST AI RMF

Govern (policy, accountability, roles, risk culture)
MAP (Intended use, stakeholders, impact, risk)
MEASURE (Testing, metrics, validation, evidence)
MANAGE (Mitigation, prioritizing, monitoring)

Govern (who is responsible for AI risk)

Example > Assigning responsibilities

AI Product Owner > Defines intended functionality and business requirements.

Data Scientist > Training data, model evaluation and limitations.

ML Engineer > Model deployment, versioning and performance monitoring.

AI-FDE > Customer integration, deployment context, operational requirements and technical evidence.

Security Engineer > Access control, infrastructure security and vulnerability management.

Safety Officer > Reviews alerts and makes operational safety decisions.

Client Management > Accepts residual risks and approves operational deployment.

The FDE should make those risks visible, translate them into engineering work, and ensure the appropriate people make and document the decisions.

Who is responsible if the AI misses a worker falling from a height?

The data scientist who trained the model?
The ML engineer who deployed it?

the AI team.

There must be defined accountability.

MAP (Understand what you are actually building)

Example > Client request > We want an AI system that detects safety violations.

is the requirement defined to the fine level?

You need to understand the actual context. > Ask the right questions
- Is the system detecting missing PPE?
- Is it detecting restricted-zone entry?
- Indoor or outdoor cameras?
- Fixed or moving cameras?
- How reliable is the network?
- What is the camera resolution?
- Workers being monitored.

Task > Industrial PPE detection

System > YOLO-based PPE detection
Input > CCTV video frames
Output > Person, helmet and safety-vest detections
Intended use > Assist safety officers with monitoring
Human involvement > Safety officer reviews alerts
Deployment > Factory premises
Key limitation > Occlusion, poor lighting, camera angles

MEASURE (How do you know the risks are real?)

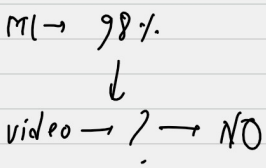
most AI engineers are familiar with > testing and evaluation.

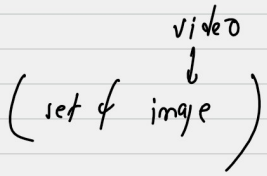
AI RMF > look beyond ordinary model metrics

YOLO > [mAP, Precision, Recall] ← image → ML

Does that prove that the system is ready for deployment? >NO

Computer vision algo (detection) >>> image -> ML --> Number of person





A missed helmet detection

why system failed :
- The worker is partially hidden.
- The camera is mounted at a steep angle.
- The lighting is poor.
- The helmet colour is unusual.
- The worker is far away from the camera.



We should measure :

Model performance > Precision, recall, mAP, confusion matrix
False negatives (Video) > How many actual safety violations were missed?
False positives(Video) > How many incorrect safety alerts were generated?
Reliability > Inference failures, service availability, latency
Environmental robustness > Performance under different lighting and camera angles

MANAGE (What will you do about the identified risks?)

This is where risk assessment becomes actual engineering work.This is where risk assessment becomes actual engineering work.

RISK(Helmet missed in low light) > **IMPACT**(Safety violation goes unnoticed) > **ENGINEERING-RESPONSE**(Collect low-light data, test again, define camera limitations)

RISK(Model update degrades performance) > **IMPACT**(Previously reliable detections become unreliable) > **ENGINEERING-RESPONSE**(Version models and require regression testing before release)